SCHOOL OF ENGINEERING AND TECHNOLOGY

A Project Report

On

[DIGITAL UNIVERSITY LIBRARY ARCHIVE CATALOG AND FUZZY KEYWORD SEARCH]

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[COMPUTER SCIENCE AND TECHNOLOGY]

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CERTIFICATE

This is to certify that the Project entitled "DIGITAL UNIVERSITY LIBRARY ARCHIVE CATALOG AND FUZZY KEYWORD SEARCH " ([Project 67]) has been successfully carried out by VARALAXMI POL- 25BBTCG070 and SABERA 25BBTCG053 in partial fulfillment of the requirements of the course **Object Oriented Programming using Java & Kotlin (4CTGC2011)** of the **Bachelor of Technology** programme in **[Computer Science and Technology]** of **CMR University**, Bengaluru during the academic year **2026–2027.** The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said course.

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ABSTRACT

The **Digital University Library Archive Catalogue and Fuzzy Keyword Search** is a digital library management and information retrieval system designed to simplify the process of searching, accessing, and managing academic resources. Traditional library search systems often require users to enter exact titles, author names, or keywords, making it difficult to find relevant resources when search terms contain spelling mistakes or variations.

The proposed system provides a centralized digital catalogue for organizing and retrieving library resources such as books, research papers, journals, academic documents, and other educational materials. It incorporates **fuzzy keyword searching** to improve search accuracy by identifying relevant records even when users enter approximate keywords, minor spelling errors, or incomplete search terms. This makes the search process more flexible, user-friendly, and efficient.

The system aims to reduce the time required to locate academic information while improving the accessibility and organization of digital library archives. It can provide features such as keyword-based searching, resource categorization, metadata management, and display of relevant search results. The implementation may use suitable database technologies and fuzzy string-matching algorithms to compare user queries with stored catalogue records.

Overall, this project aims to develop an efficient and accessible digital library archive catalogue that enhances information retrieval through fuzzy keyword search techniques. It can benefit students, faculty members, researchers, and librarians by making academic resources easier to discover and manage.

Keywords:
Digital library,University library,Archive Catalog,Fuzzy keyword search, information,Retrival,Digital Archives,Search optimization, Meta data management,String matching,Academic resourse management.

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1. INTRODUCTION

In the modern digital era, libraries play an important role in providing access to academic information and educational resources. Traditional library systems often require users to search for books, journals, research papers, and other documents using exact titles, author names, or keywords. This process can become difficult when users do not know the exact information or enter keywords with spelling mistakes.

The **Digital University Library Archive Catalogue and Fuzzy Keyword Search** project aims to develop a digital platform that helps users search for and retrieve academic resources efficiently. The system maintains a centralized catalogue of library resources and allows users to search using keywords related to titles, authors, subjects, and other available details.

The main feature of this project is fuzzy keyword searching, which helps identify relevant resources even when the search query contains minor spelling errors, incomplete words, or variations in the entered terms. This makes the search process more flexible and user-friendly than a system that relies only on exact keyword matching.

The proposed system aims to improve information accessibility, reduce the time required to find academic materials, and simplify the management of digital library archives. It can be useful for students, faculty members, researchers, and librarians who need quick and convenient access to educational resources.

1.1 Problem Statement

Traditional library catalogue systems may require users to enter exact keywords, titles, or author names to retrieve relevant information. If a user enters a spelling mistake, an incomplete keyword, or a slightly different search term, the system may fail to display the required resource.

Searching through large collections of academic documents can also be time-consuming when resources are not organized or retrieved efficiently. These limitations make it difficult for students, faculty members, and researchers to locate relevant information quickly.

Therefore, there is a need for a digital university library archive catalogue that organizes academic resources and supports fuzzy keyword searching. The proposed system aims to improve search flexibility by identifying relevant records based on approximate keyword matches, making information retrieval more efficient, convenient, and accessible.

1.2 Objectives

The main objectives of the Digital University Library Archive Catalogue and Fuzzy Keyword Search project are:

1. **To develop a digital library catalogue** for organizing and managing academic resources such as books, journals, research papers, and documents.
2. **To implement fuzzy keyword search** that helps users find relevant resources even when search terms contain minor spelling errors or variations.
3. **To improve information retrieval** by providing relevant search results based on user-entered keywords.
4. **To reduce search time** and make it easier for users to locate required academic materials.
5. **To organize library resources systematically** using details such as title, author, subject, and other relevant metadata.
6. **To provide a user-friendly search interface** that allows students, faculty members, and researchers to search for resources conveniently.
7. **To improve accessibility to digital archives** by simplifying the process of discovering and retrieving academic information.
8. **To enhance search accuracy and flexibility** compared with systems that depend entirely on exact keyword matching.

1.3 Scope

The scope of the **Digital University Library Archive Catalogue and Fuzzy Keyword Search** project is to develop a digital platform that simplifies the organization, searching, and retrieval of academic resources. The system focuses on providing an efficient and user-friendly way to access library information through keyword-based and fuzzy keyword searching.

The system can maintain a catalogue of academic resources, including books, journals, research papers, project reports, and other digital documents. Users can search for resources using details such as titles, author names, subjects, or relevant keywords. The fuzzy keyword search feature helps identify potentially relevant records even when users enter minor spelling mistakes, incomplete words, or approximate search terms.

The project also focuses on improving the speed and convenience of information retrieval, reducing the effort required to locate academic materials, and making digital library archives easier to explore. It can be useful for students, faculty members, researchers, and librarians in academic institutions.

The system can be extended in the future by incorporating advanced search filters, document previews, downloadable resources, user authentication, recommendation systems, and support for larger digital collections. The actual scope depends on the features implemented in the final project.

Overall, this project aims to make academic information more accessible and improve the search experience through fuzzy keyword matching and organized digital catalogue management.

1.4 Advantages of the Proposed System

The proposed Digital University Library Archive Catalogue and Fuzzy Keyword Search system offers several advantages over traditional library searching methods.

1. **Faster Information Retrieval:** Helps users find relevant academic resources quickly, reducing the time spent searching for information.
2. **Fuzzy Keyword Searching:** Identifies potentially relevant results even when users enter minor spelling mistakes or approximate keywords
3. **Improved Search Flexibility:** Allows users to search using titles, author names, subjects, and other available keywords without always requiring exact matches.
4. **Centralized Digital Catalogue:** Organizes library resources in a single catalogue, making information easier to manage and retrieve.
5. **User-Friendly Interface:** Provides a convenient search process for students, faculty members, researchers, and librarians.
6. **Reduced Manual Effort:** Minimizes the need to search through physical catalogues or browse large collections manually.
7. **Better Resource Discovery:** Helps users discover relevant books, journals, research papers, and other academic materials.
8. **Efficient Resource Organization:** Supports systematic organization of academic resources using details such as titles, authors, subjects, and metadata.
9. **Time and Effort Saving:** Simplifies the process of locating academic information and improves the overall search experience.
10. **Future Expandability:** The system can be enhanced with advanced filters, document previews, user authentication, and additional search features as required.

In conclusion, the proposed system improves the convenience, flexibility, and efficiency of academic information retrieval through digital catalogue management and fuzzy keyword searching.

2. REQUIREMENTS

2.1 Functional Requirements

Functional requirements describe the main operations that the Digital University Library Archive Catalogue and Fuzzy Keyword Search system must perform.

1. **User Search:** The system shall allow users to search for library resources using keywords.

2. **Fuzzy Keyword Search:** The system shall identify approximate keyword matches, including minor spelling mistakes and variations.

3. **Resource Retrieval:** The system shall retrieve relevant library records based on the user's search query.

4. **Catalogue Management:** The system shall store and organize details of books, journals, research papers, and other academic resources.

5. **Search Results Display:** The system shall display matching resources with details such as title, author, subject, and description, where available.

6. **Resource Addition:** Authorized users shall be able to add new catalogue records if administrative functionality is implemented.

7. **Resource Updating and Deletion:** Authorized users shall be able to update or delete existing records if these features are supported.

8. **User-Friendly Interface:** The system shall provide a simple interface for entering search terms and viewing results.

9. **Database Interaction:** The system shall retrieve catalogue information from the database and provide relevant results to the user.

2.2 Hardware and Software Requirements

Software Requirments:

1. **Operating System:** Windows, Linux, or macOS.
2. **Programming Language:** Python or Java, depending on the implementation.
3. **Frontend Technologies:** HTML, CSS, and JavaScript, if the application is web-based.
4. **Backend Framework:** Flask, Django, or another suitable framework, if required.
5. **Database Management System:** MySQL, SQLite, or another suitable database.
6. **Fuzzy Search Technique:** Levenshtein distance or another fuzzy string-matching algorithm.
7. **Development Environment:** Visual Studio Code, PyCharm, or another suitable IDE.
8. **Web Browser:** Google Chrome, Microsoft Edge, or Mozilla Firefox, if the application is web-based.

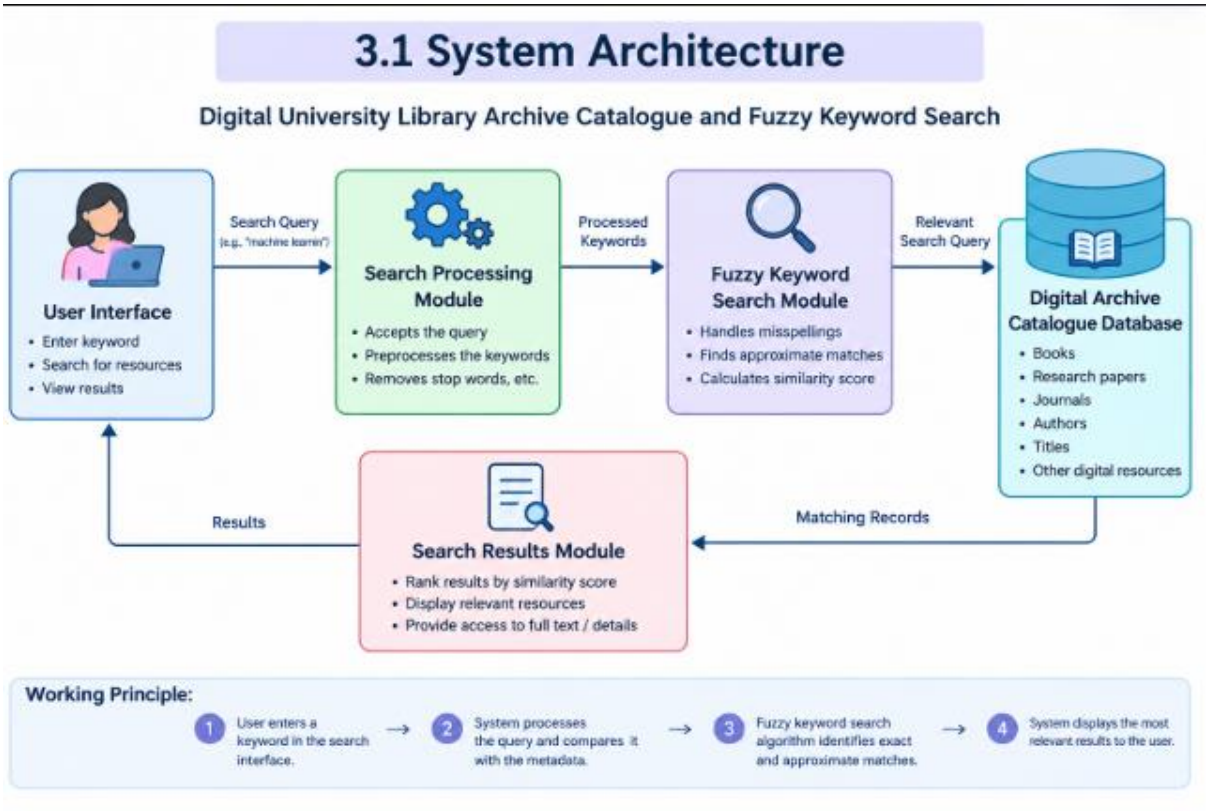
Hardware Requirements

The following hardware components are required to develop, run, and test the proposed system.

1. **Processor:** Intel Core i3 or equivalent processor and above.
2. **RAM:** Minimum 4 GB.
3. **Hard Disk:** Minimum 10 GB of available storage space.
4. **Computer System:** Desktop computer or laptop.
5. **Input Devices:** Keyboard and mouse for user interaction.
6. **Output Device:** Monitor or laptop display to view the application and search results.
7. **Internet Connection:** Required for accessing online resources or remote databases, if applicable.

3. SYSTEM DESIGN

3.1 System Architecture



The system architecture describes the overall structure of the Digital University Library Archive Catalogue and Fuzzy Keyword Search system. It explains how users interact with the system to search, retrieve, and access digital library resources.

The proposed system consists of the following modules:

1. User Interface: Allows users to enter keywords and search for digital library resources.
2. Search Processing Module: Accepts the search query and processes the entered keywords.
3. Fuzzy Keyword Search Module: Identifies relevant resources even when users enter misspelled words or keywords that do not exactly match the stored records.
4. Digital Archive Catalogue Database: Stores information about books, research papers, journals, authors, titles, and other digital resources.
5. Search Results Module: Displays relevant resources based on the user's query and similarity score.

Working Principle:

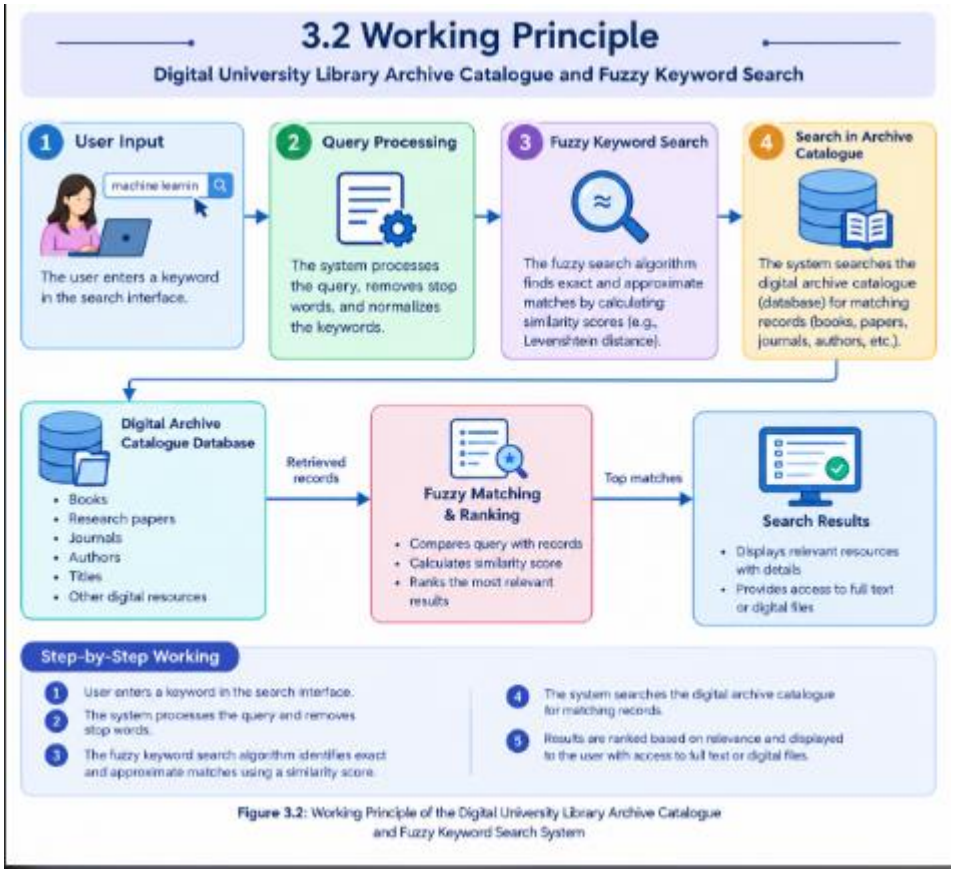


Figure 3.2: Working Principle of the Digital University Library Archive Catalogue and Fuzzy Keyword Search System

The user enters a keyword in the search interface. The system processes the query and compares it with the metadata stored in the digital archive catalogue. The fuzzy keyword search algorithm

identifies exact and approximate matches. Finally, the system displays the most relevant results to the user.

3.2 UML Class Diagram

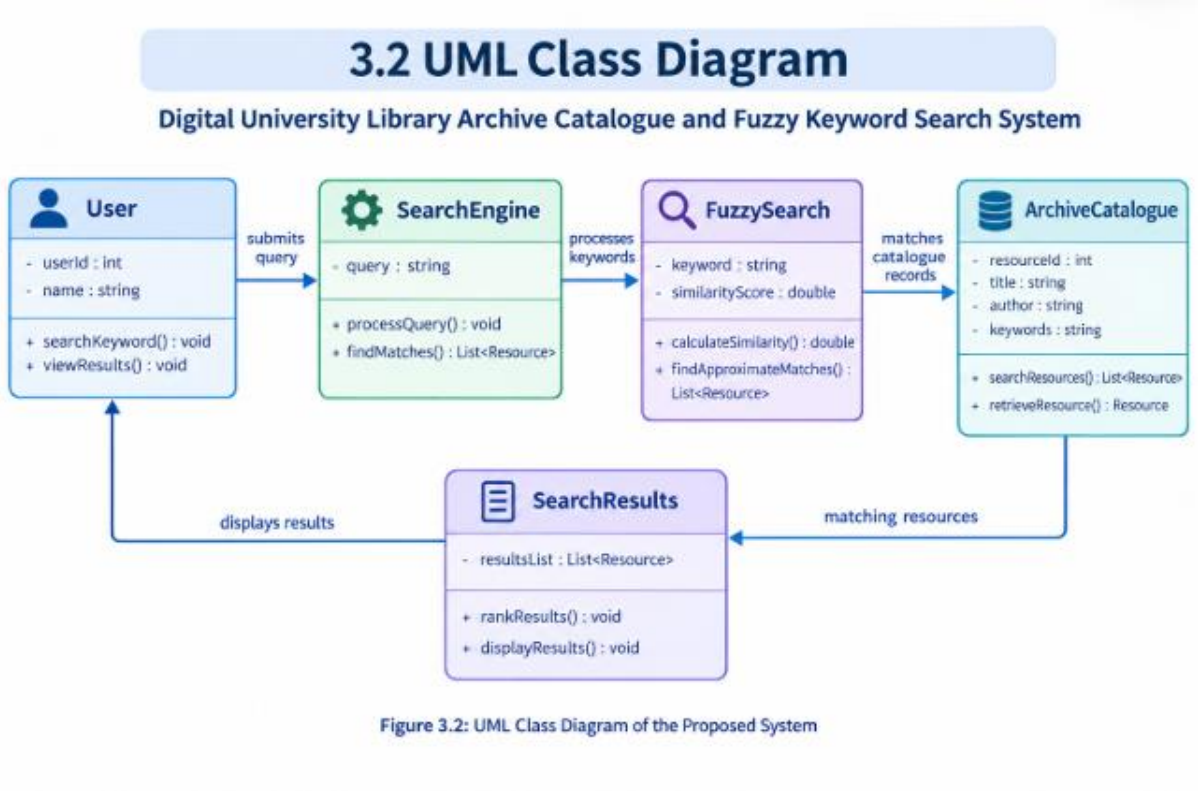


Figure 3.2: UML Class Diagram of the Proposed System

The UML class diagram represents the main classes of the proposed system, including User, SearchEngine, FuzzySearch, ArchiveCatalogue, and SearchResults. Each class contains its attributes and methods. These classes interact to process user queries, identify exact and approximate keyword matches, retrieve relevant digital resources, and display the search results efficiently.

4. IMPLEMENTATION

4.1 OOP Concepts Used

Object-Oriented Programming (OOP) concepts help organize the system into classes and objects, making the application easier to develop, maintain, and extend. The main OOP concepts that can be applied to the proposed system are:

1. Class and Object

A class defines the properties and methods of an entity, while an object is an instance of a class. In this project, classes such as `User`, `SearchEngine`, `FuzzySearch`, and `ArchiveCatalogue` can be used to organize the system's functionality.

2. Encapsulation

Encapsulation combines data and methods within a single class. For example, the `ArchiveCatalogue` class can store resource details such as title, author, and keywords and provide methods to retrieve the information.

3. Abstraction

Abstraction hides internal implementation details and provides only the necessary functionality to the user. In this project, users enter keywords and receive search results without needing to understand how the fuzzy search algorithm works internally.

4. Inheritance

Inheritance allows one class to acquire properties and methods from another class. For example, specialized resource classes such as `Book` and `Journal` could inherit common attributes from a general `Resource` class.

5. Polymorphism

Polymorphism allows the same method name or interface to perform different operations depending on the object. For example, different resource types could implement their own versions of a `displayDetails()` method.

These OOP concepts help improve code reusability, modularity, readability, and maintainability.

Note: Include only the OOP concepts actually implemented in your project code.

4.2 Code Snippets

1. Resource Class — Classes and Objects

class Resource {

String title;

String author;

String keywords;

```
Resource(String title, String author, String keywords) {
```

```
this.title = title;
```

```
        this.author = author;
```

```
        this.keywords = keywords;
```

```
}
```

```
void displayDetails() {
```

```
System.out.println("Title: " + title);
```

```
System.out.println("Author: " + author);
```

```
System.out.println("Keywords: " + keywords);
```

```
}
```

```
}
```

Explanation: The `Resource` class represents a digital library resource. It stores the title, author, and keywords and displays the resource details using the `displayDetails()` method.

2. Fuzzy Keyword Search

```
import java.util.*;
```

class FuzzySearch {

```
static int editDistance(String s1, String s2) {
```

```
int[][] dp = new int[s1.length() + 1][s2.length() + 1];
```

```
for (int i = 0; i <= s1.length(); i++)
```

```
dp[i][0] = i;
```

```
for (int j = 0; j <= s2.length(); j++)
```

```
dp[0][j] = j;
```

```
for (int i = 1; i <= s1.length(); i++) {
```

```
for (int j = 1; j <= s2.length(); j++) {
```

```
int cost = s1.charAt(i - 1) == s2.charAt(j - 1)
```

```
? 0 : 1;
```

```
dp[i][j] = Math.min(
```

```
Math.min(dp[i - 1][j] + 1,
```

```
                dp[i][j - 1] + 1),
```

```
dp[i - 1][j - 1] + cost
```

```
);
```

```
}
```

```
}
```

```
    return dp[s1.length()][s2.length()];
```

```
}
```

```
    static boolean matches(String query, String keyword) {
```

```
query = query.toLowerCase();
```

```
keyword = keyword.toLowerCase();
```

```
int distance = editDistance(query, keyword);
```

```
        return distance <= 2;
```

```
}
```

```
}
```

xplanation: This code implements fuzzy keyword matching using the Levenshtein edit distance algorithm. It calculates the minimum number of insertions, deletions, or substitutions required to transform one string into another. A distance of two or less is considered a match in this example.

3. Archive Catalogue Search

```
import java.util.*;
```

class ArchiveCatalogue {

```
ArrayList<Resource> resources = new ArrayList<>();
```

```
    void addResource(Resource resource) {
```

```
resources.add(resource);
```

```
}
```

```
void searchResources(String query) {
```

```
boolean found = false;
```

```
for (Resource r : resources) {
```

```
if (FuzzySearch.matches(query, r.title)
```

```
            || FuzzySearch.matches(query, r.keywords)) {
```

```
        r.displayDetails();
```

```
        System.out.println("----------------");
```

```
found = true;
```

```
}
```

```
}
```

```
if (!found) {
```

```
System.out.println("No matching resources found.");
```

```
}
```

```
}
```

```
}
```

Explanation: The `ArchiveCatalogue` class stores resource objects in an `ArrayList`. It searches titles and keywords using the fuzzy matching method and displays matching records.

4. Main Class — Executing the System

```
public class Main {
```

```
public static void main(String[] args) {
```

```
ArchiveCatalogue catalogue = new ArchiveCatalogue();
```

```
catalogue.addResource(new Resource(
```

"Machine Learning",

"Tom Mitchell",

"Artificial Intelligence Data"

```
));
```

```
catalogue.addResource(new Resource(
```

"Computer Networks",

"Andrew Tanenbaum",

"Networking Internet"

```
));
```

```
catalogue.searchResources("Machin Learning");
```

```
}
```

```
}
```

Explanation: The `Main` class creates the archive catalogue, adds sample resources, and searches using a keyword containing a spelling mistake. The fuzzy search attempts to identify the corresponding resource.

5. RESULTS

5.1 Execution Snapshots

This section presents the execution snapshots of the Digital University Library Archive Catalogue and Fuzzy Keyword Search system. The screenshots demonstrate the execution of the Java program and its ability to search digital library resources using keywords.

Figure 5.1: Main Program Execution

Insert a screenshot of the Java program running in your IDE or terminal.

Figure 5.2: Keyword Search Operation

Insert a screenshot showing the user entering a keyword and the system searching the catalogue.

Figure 5.3: Fuzzy Keyword Search Result

Insert a screenshot showing the system identifying a relevant resource when the entered keyword contains a spelling mistake.

Figure 5.4: Display of Search Results

Insert a screenshot showing the matching resource details, including title, author, and keywords.

Result: The execution snapshots illustrate the main operations of the proposed system, including resource searching, fuzzy keyword matching, and displaying relevant results. These screenshots should be captured from the actual Java program to demonstrate its functionality.

Important: Run your Java program and insert the real output screenshots under each figure. Do not use sample screenshots as proof of actual execution.

6. CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The Digital University Library Archive Catalogue and Fuzzy Keyword Search system provides an efficient approach to searching and retrieving digital library resources. The system is designed to help users find books, research papers, journals, and other academic materials using keywords. By incorporating fuzzy keyword matching, it can identify approximate matches and help users find relevant resources even when their search queries contain minor spelling mistakes.

The system uses Java and Object-Oriented Programming (OOP) concepts to organize resource information and search operations. It improves the flexibility of keyword searching and makes academic information easier to locate. The proposed system can be useful for students, researchers, faculty members, and other users who need quick access to digital library resources.

In conclusion, the project demonstrates how fuzzy keyword searching can improve resource discovery and provide a more convenient search experience in a digital university library.

6.2 Future Scope

The system can be enhanced in several ways in the future:

1. Database Integration: Connect the application to a database such as MySQL to store and manage large collections of digital resources.

2. Advanced Search: Add filters for author, title, publication year, subject, and resource type.

3. Improved Fuzzy Matching: Implement advanced algorithms to improve the accuracy of approximate keyword matching.

4. Graphical User Interface: Develop a user-friendly interface using Java Swing or JavaFX.

5. Online Access: Make the system available through a web application so users can access library resources from different locations.

6. User Authentication: Introduce login facilities for students, faculty, and library administrators.

7. Relevance-Based Ranking: Rank search results according to their relevance to the entered keywords.

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